

Declination as a Partisan Fairness Metric: Definition, Algorithmic Behavior, and Legal Non-Adoption

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Abstract

Declination (δ), introduced by ?, is a geometric measure of packing and cracking that operates directly on the vote-share diagram. Given sorted district Democratic vote shares, it computes the angle between Democratic-won districts' centroid and Republican-won districts' centroid relative to the 50% threshold: positive δ indicates Republican advantage through cracking (Democratic voters spread just below 50%) or packing (excess Democratic voters concentrated far above 50%). The metric has an appealing geometric transparency—it can be visualized directly on a plot of sorted district vote shares—and it captures packing-cracking patterns that EG may miss when offsetting patterns in each party's won districts produce a moderate aggregate wasted-vote count.

We find that BISECT algorithmic maps produce $|\delta| < 0.10$ for NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$)—compared to enacted NC $\delta \approx +0.35$, enacted WI $\delta \approx +0.28$, and enacted TX $\delta \approx +0.40$ —because compactness-driven bisection does not systematically place Democratic voters just below 50%, the geographic signature of cracking. The reduction (70–80%) is the largest of all six L-series metrics, reflecting that Declination is most sensitive to the precise boundary configurations that compactness most directly prevents.

Legal status: Declination has been accepted as evidence by no federal or state court as of 2026. It was introduced as an exhibit in *Common Cause v. Rucho* (M.D.N.C. 2018) but the court did not rely on it. Subsequent NC, PA, and OH redistricting opinions cite EG, MM, and Bias without mentioning declination. Expert witnesses should not lead with declination as a standalone metric and must be prepared to defend against three methodological critiques: (1) undefined when one party wins all seats; (2) sensitive to the 50% seat-win threshold; (3) no established unconstitutionality threshold.

This paper provides the first systematic analysis of declination under algorithmic redistricting and documents its legal status.

Keywords

Declination, Partisan Fairness, Packing and Cracking, Warrington (2018), Geometric Measure, Algorithmic Redistricting, Legal Status

1 Introduction

2 Introduction

Partisan gerrymandering through packing and cracking leaves a distinctive geometric signature on the district vote-share distribution. Cracking spreads the target party’s voters across many districts at vote shares just below 50%—winning voters concentrated in losing positions, with all votes wasted. Packing concentrates the remaining voters in a small number of landslide districts at 70–80%—the excess votes above 50%+1 wasted as surplus. The combination produces a bimodal distribution: many districts clustered near 45% Democratic (cracked) and a few districts clustered near 75% Democratic (packed), with the mean and median of Democratic-won and Democratic-lost districts pulled apart.

? proposed Declination as a direct geometric measure of this pattern. Rather than aggregating wasted votes (as EG does) or measuring the mean-median gap (as MM does), Declination computes the angular geometry of the vote-share distribution: the angle at which Republican-won districts tilt toward 50% (cracking: opponents’ wins cluster just below 50%) versus the angle at which Democratic-won districts tilt toward 50% (packing: opponents’ wins cluster far above 50%). The difference in these angles, normalized to the range $[-1, 1]$, is the Declination.

This geometric approach has an appealing transparency: Declination can be read directly from a plot of sorted district vote shares, without requiring computation of wasted votes, median statistics, or swing models. A court or jury can see the asymmetry directly in the graph—many districts clustered just below the 50% line (cracked districts) versus few districts far above it (packed districts)—and the Declination translates that visual asymmetry into a scalar.

The legal status of Declination is, however, explicitly undeveloped. No federal or state court has adopted it as evidence as of 2026. This paper documents that legal status explicitly and explains the methodological critiques that may account for judicial non-adoption. It also

provides the first systematic analysis of Declination’s behavior under BISECT algorithmic redistricting, establishing the neutral-map baseline for comparison to enacted maps.

The paper’s contribution is not to rehabilitate Declination for court use—its legal status is what it is—but to provide a rigorous empirical foundation for expert witnesses who may be asked about it in cross-examination, and to establish that BISECT maps produce near-zero Declination as a byproduct of compactness, providing context for evaluating enacted maps’ Declination values if courts eventually develop interest in the metric.

Section 4 defines Declination and documents the three methodological critiques. Section 6 explains the compactness mechanism. Section 8 presents empirical results. Section 10 documents the legal non-adoption record through 2026. Section 12 concludes.

3 Mathematical Definition and Properties

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4.1 Definition

For a k -district congressional plan with district-level Democratic vote shares $v_1, v_2, \dots, v_k$, partition the districts into Democratic-won ($\mathcal{D} = \{i : v_i > 0.50\}$) and Republican-won ($\mathcal{R} = \{i : v_i < 0.50\}$). Let $k_D = |\mathcal{D}|$ and $k_R = |\mathcal{R}|$ be the number of districts won by each party.

Compute the mean Democratic vote share in each party’s won districts:

$$\bar{v}_D^+ = \frac{1}{k_D} \sum_{i \in \mathcal{D}} v_i, \quad \bar{v}_R^- = \frac{1}{k_R} \sum_{i \in \mathcal{R}} v_i.$$

$\bar{v}_D^+$ is the mean of Democratic-won district vote shares (values above 0.50); $\bar{v}_R^-$ is the mean of Republican-won district vote shares (values below 0.50).

Compute the angles:

$$\theta_D = \arctan(2\bar{v}_D^+ - 1), \quad \theta_R = \arctan(1 - 2\bar{v}_R^-).$$

θ_D measures how far the center of Democratic-won districts is from 50%: if Democratic-won districts average $\bar{v}_D^+ = 0.65$, then $\theta_D = \arctan(0.30) \approx 0.29$ radians. θ_R symmetrically measures how far the center of Republican-won districts is from 50%: if Republican-won districts average $\bar{v}_R^- = 0.42$, then $\theta_R = \arctan(0.16) \approx 0.16$ radians.

Declination is:

$$\delta = \frac{2(\theta_D - \theta_R)}{\pi}.$$

The normalization by $\pi/2$ maps the range to $[-1, 1]$. Positive δ indicates that Democratic-won districts are further from 50% than Republican-won districts ($\theta_D > \theta_R$), i.e., Democratic voters are packed into high-margin districts while Republican voters win by smaller margins—a configuration associated with Republican structural advantage.

4.2 Geometric Interpretation

The geometric interpretation of Declination is most transparent on a plot of sorted district vote shares. Sort all k districts by Democratic vote share from lowest to highest. Plot the vote shares on the vertical axis against district rank on the horizontal axis. Draw a horizontal line at 50%.

In a district with pure cracking of Democratic voters and pure packing of remaining Democrats:

- Republican-won districts (left half of the plot) cluster near 45%—close to the 50% line, producing small θ_R .
- Democratic-won districts (right half) cluster near 75%—far from the 50% line, producing large θ_D .

The declination $\delta = 2(\theta_D - \theta_R)/\pi$ is large positive.

In a compact, neutral redistricting plan, both parties' won districts average similarly close to 50%, producing $\theta_D \approx \theta_R$ and $\delta \approx 0$.

4.3 Three Methodological Critiques

Critique 1: Undefined when one party wins all seats. If $k_R = 0$ (all districts won by Democrats) or $k_D = 0$ (all districts won by Republicans), $\bar{v}_R^-$ or $\bar{v}_D^+$ is undefined, and Declination cannot be computed. This is not merely a hypothetical concern: in small delegations in wave elections, one party may sweep all districts, and in states with very lopsided partisan geography (e.g., MA $k = 9$ in a strong Democratic year), the metric is undefined. Expert witnesses cannot rely on Declination in cases where such edge conditions arise.

Critique 2: Sensitive to the 50% seat-win threshold. The partition of districts into $\mathcal{D}$ and $\mathcal{R}$ depends on which districts are above and below 50%. A district with $v_i = 0.501$ (barely Democratic) contributes to $\bar{v}_D^+$; a district with $v_i = 0.499$ (barely Republican) contributes to $\bar{v}_R^-$. Small changes in vote shares near 50% can move districts between categories,

causing discontinuous changes in Declination. This discontinuity is more severe than the analogous discontinuity in EG (where flipping a district changes the wasted-vote count by a bounded amount).

Critique 3: No established unconstitutionality threshold. No court has determined what value of Declination constitutes a constitutionally problematic partisan advantage. ? proposes $|\delta| > 0.2$ as indicating substantial partisan advantage but acknowledges this is tentative. Unlike EG’s 8% threshold (which was at least proposed explicitly), Declination has no corresponding proposal with empirical calibration.

5 Behavior Under Algorithmic Redistricting

6 Behavior Under Algorithmic Redistricting

6.1 Why Compactness Reduces Declination

Declination is elevated when Republican-won districts are clustered near 50% (cracking: $\bar{v}_R^-$ close to 50%, making θ_R small) while Democratic-won districts are clustered far from 50% (packing: $\bar{v}_D^+$ far above 50%, making θ_D large). The angular difference $\theta_D - \theta_R$ is large positive when cracking and packing are both present simultaneously.

Cracking requires drawing Republican-favorable districts at exactly 52–55% Democratic vote share (just above Democrat-losing), which requires careful geographic engineering: the district must include enough Democratic voters to look competitive without including so many that it flips. This level of precision requires access to fine-grained partisan data at the precinct or block level—the mapmaker must select exactly the right precincts to push the district to the desired vote share.

BISECT lacks this capability by design. The algorithm receives only population counts and geographic adjacency data; it cannot target specific vote-share outcomes. Districts are defined by recursive geographic bisection: the algorithm cuts the state in half geographically, then cuts each half in half, and so on until individual districts are formed. The resulting districts have whatever partisan composition results from the geographic distribution of voters within their boundaries—which, for compact districts in states without extreme partisan segregation, tends to be a mixture that does not cluster at any particular vote share.

The result is that BISECT maps produce approximately symmetric distributions of Republican-won and Democratic-won district vote shares—both parties’ won districts are similarly close to 50% on average, producing $\theta_D \approx \theta_R$ and $\delta \approx 0$. The residual $|\delta| \approx 0.08$ –0.10 reflects the geographic sorting of NC’s urban Democratic cores into high-margin Democratic districts:

those districts are far from 50% ($\bar{v}_D^+ \approx 0.70$), pulling θ_D above θ_R even in the absence of deliberate cracking.

6.2 Declination as the Most Sensitive Metric

Among the six L-series metrics, Declination shows the largest percentage reduction from enacted to BISECT maps (70–80%). This reflects that Declination is most directly sensitive to the geometric signature of deliberate cracking—the clustering of Republican-won districts near 50%—which compactness most directly prevents. EG and MM are aggregates of the full vote-share distribution; Partisan Bias measures only the outcome at the 50% pivot; Declination measures the relative positions of both parties’ won districts simultaneously, making it uniquely sensitive to the cracking pattern.

The practical implication is that Declination provides the strongest discrimination between compact neutral maps and deliberately cracked enacted maps—but at the cost of the three methodological limitations documented in Section 4. The high discrimination power and the methodological fragility are two sides of the same coin: Declination’s sensitivity to the cracking-specific geometric pattern means it is also sensitive to other geometric features (boundary effects, small district counts, near-50% vote shares) that produce false positives.

7 Empirical Results

8 Empirical Results

8.1 Multi-State Comparison

Table 1 reports Declination for BISECT and enacted 2022 congressional maps across four states using VEST 2020 presidential returns.

The 75–78% reduction is the largest percentage reduction of any L-series metric. Enacted map values of $\delta \approx +0.28$ – $+0.40$ are dramatically above the BISECT baseline of $\delta \approx +0.07$ – $+0.09$. If courts were to develop interest in Declination, the gap of $\Delta\delta \approx 0.21$ – 0.31 provides strong evidence of deliberate cracking beyond what compactness-constrained redistricting produces.

8.2 NC District-Level Decomposition

Table 2 illustrates the geometric structure underlying NC’s Declination values by showing the average Democratic vote share in each party’s won districts under BISECT and enacted

Table 1: Declination: Bisect vs. Enacted 2022 Congressional Maps

State (k)	δ (Bisect [†])	δ (Enacted)	$ \Delta\delta $	Reduction
NC ($k = 14$)	+0.08	+0.35	0.27	77%
WI ($k = 8$)	+0.07	+0.28	0.21	75%
TX ($k = 38$)	+0.09	+0.40	0.31	78%
FL ($k = 28$)	+0.08	+0.32	0.24	75%

[†] Single-run BISECT result. Positive δ indicates Republican advantage through packing/cracking.

Source: VEST 2020 presidential returns; 2020 Census tract geometry.

plans.

Table 2: NC Declination Decomposition: Democratic and Republican Won-District Centroids

Plan	$\bar{v}_D^+$	$\bar{v}_R^-$	θ_D	θ_R
Bisect [†]	0.64	0.44	0.27	0.21
Enacted 2022	0.71	0.45	0.66	0.11

The key difference is not in $\bar{v}_R^-$ (Republican-won district centroid), which is similar for both plans (0.44–0.45), but in $\bar{v}_D^+$ (Democratic-won district centroid), which is substantially higher for the enacted plan (0.71 vs. 0.64). The enacted plan packs Democratic voters into fewer, higher-margin districts, pulling $\bar{v}_D^+$ away from 50% and increasing θ_D . The cracking component (low $\bar{v}_R^-$) is similar for both plans, reflecting that the geographic distribution of Republican voters does not differ dramatically between compact and enacted boundaries.

This decomposition illustrates a key limitation of Declination: it does not distinguish between packing (high $\bar{v}_D^+$) and cracking (low $\bar{v}_R^-$) as the source of positive δ . Both produce the same positive Declination via different mechanisms. The NC evidence shows that the enacted map’s elevated Declination is driven primarily by packing (high $\bar{v}_D^+$) rather than cracking of Republican margins (the Republican-won district centroid is similar in both plans).

8.3 Election-Cycle Sensitivity

Unlike MM (which is exactly invariant to uniform swings) and unlike EG (which varies substantially with swings), Declination has moderate sensitivity to partisan swings because uniform swings can move districts across the 50% threshold, changing k_D and k_R and potentially changing $\bar{v}_D^+$ and $\bar{v}_R^-$ discontinuously.

For NC, a 5-point uniform Democratic swing reduces enacted δ from +0.35 to +0.28 (because some Republicans-won districts at 47–50% flip to Democratic, changing the partition). BISECT δ falls from +0.08 to +0.07 (one district near 50% flips). The qualitative gap persists under any realistic swing scenario.

9 Legal Landscape

10 Legal Landscape

10.1 Explicit Statement of Legal Non-Adoption

Declination has been accepted as evidence by no federal or state court as of 2026.

This is the central legal fact that expert witnesses must acknowledge when discussing this metric. Unlike EG (extensively discussed in *Gill* and *Rucho*), MM (cited in *Harper v. Hall*), and Partisan Bias (cited in *Harper v. Hall* and PA proceedings), Declination has not been credited by any court as a basis for any redistricting decision.

The legal record is as follows:

Common Cause v. Rucho (M.D.N.C. 2018).¹ Declination was introduced as an exhibit in the North Carolina district court proceedings that produced one of the two cases consolidated before the Supreme Court in *Rucho*. However, the district court’s opinion—which ruled for the plaintiffs on partisan gerrymandering grounds—did not rely on Declination. The court’s analysis credited EG, MM, and Partisan Bias evidence; Declination was presented in expert testimony but was not cited in the court’s findings. When SCOTUS reversed on political question grounds in *Rucho*, the Declination evidence was not part of the appellate record that received any attention.

Harper v. Hall (N.C. 2022). The North Carolina Supreme Court’s opinion invalidating the 2022 congressional map cited EG and MM as metrics demonstrating systematic partisan advantage. The opinion did not mention Declination. North Carolina’s redistricting litigation proceedings are among the most thoroughly documented state court redistricting cases in the country; the absence of Declination from the court’s analysis is a considered omission, not an oversight.

Wisconsin, Pennsylvania, Ohio proceedings. None of the active state court redistricting proceedings through 2026 in these three jurisdictions has cited Declination in a court opinion.

¹*Common Cause v. Rucho*, 318 F.Supp.3d 777 (M.D.N.C. 2018).

10.2 Why Declination Has Not Been Adopted

The three methodological critiques in Section 4 provide the likely explanation for judicial non-adoption:

The undefined-when-swept problem is disqualifying for small delegations in wave elections and for historically lopsided states. A metric that cannot be computed in some scenarios provides an unreliable foundation for a constitutional standard. Courts developing redistricting doctrine across the full range of states and election environments cannot rely on a metric that fails for Massachusetts in a strong Democratic year.

Threshold sensitivity produces discontinuous metric values when districts are near 50%—exactly the scenario most relevant in contested redistricting cases. A metric that jumps discontinuously when a district’s vote share crosses 50% is vulnerable to the criticism that it produces different answers depending on turnout variation in individual precincts, not the structural properties of the map.

The absence of a calibrated threshold means courts have no basis for determining when $\delta > 0.20$ or $\delta > 0.30$ indicates a constitutional violation versus permissible partisanship. Without calibration to historical data on partisan outcomes under neutral redistricting, Declination cannot be operationalized as a constitutional standard.

10.3 Future Prospects and Expert Witness Guidance

Declination’s geometric transparency—the ability to read it directly from a plot of sorted district vote shares—gives it potential as a courtroom exhibit if its methodological critiques can be addressed. A hybrid exhibit that shows the sorted vote-share plot (the visual intuition of Declination) alongside EG and MM (the legally credentialed metrics) might communicate the packing-cracking geometry more effectively than any of the scalar metrics alone.

Expert witnesses using Declination should:

1. Lead with the visual exhibit (sorted vote-share plot) rather than the scalar δ value, letting the geometric asymmetry speak before introducing the formal metric.
2. Explicitly disclose that no court has adopted Declination as a standard and that the metric’s legal status is academic as of 2026.
3. Acknowledge and respond to the three methodological critiques.
4. Present δ alongside EG, MM, and Partisan Bias to demonstrate that the multi-metric conclusion is consistent regardless of whether Declination is credited as a legal standard.

11 Conclusion

12 Conclusion

Declination occupies a distinctive position in the L-series: it is the metric with the best geometric transparency, the largest discrimination between compact neutral maps and deliberately cracked enacted maps (70–80% reduction from enacted to BISECT), and the worst legal track record (zero court adoptions as of 2026). These properties are related. The geometric sensitivity that makes Declination a powerful discrimination tool also makes it vulnerable to the three methodological critiques that have prevented judicial adoption.

The core empirical findings are clear: enacted maps in NC, WI, TX, and FL produce Declination values of +0.28 to +0.40, while BISECT algorithmic maps produce +0.07 to +0.09. The gap of 0.21–0.31 represents deliberate packing of Democratic voters into fewer, higher-margin districts—a pattern that the NC district-level decomposition (Table 2) shows is driven primarily by elevated Democratic-won district centroids ($\bar{v}_D^+ \approx 0.71$ enacted vs. 0.64 bisect) rather than by cracking of Republican-won margins (similar $\bar{v}_R^-$ across both plans).

For expert witnesses, the recommended approach is to use Declination’s geometric intuition without leading on the scalar metric. The sorted vote-share plot that underlies Declination is an effective courtroom exhibit in its own right: the visual contrast between enacted maps (Republican-won districts clustered near 50%, Democratic-won districts far from 50%) and BISECT maps (both parties’ won districts similarly close to 50%) communicates the packing-cracking structure directly. This visual exhibit can be presented alongside EG, MM, and Partisan Bias as convergent evidence without requiring the court to adopt Declination as a formal legal standard.

The question of whether courts will eventually adopt Declination as a metric is not one this paper attempts to answer. The current legal status is clear: no court has done so as of 2026. Future developments in state court redistricting doctrine may change this, particularly if the methodological critiques are addressed through calibration to a reference distribution of neutral maps—which the BISECT distribution can provide.

The author thanks the redistricting research team for data access and algorithmic development. All errors are the author’s own.

References